

平时(作业+点名)30% + 随堂测试 20% + 线上(每周周日) + 期末50% (折合60)

概率论 (1.5)

秋-周 A3 A5 A8 A12 A14 B1

A3 (1) $S = \{12, 13, 21, 23, 31, 32\}$

(2) $S = \{11, 12, 13, 21, 22, 23, 31, 32, 33\}$

② (3) $S = \{x \mid x \in \mathbb{N}\}$

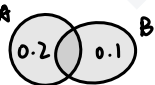
(4) $S = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq x+y \leq 1\}$

A5. $P(A \cup B) = 0.9 = P(A) + P(B) - P(AB)$

(1) $P(B \cap \bar{A}) = 0.4 = P(B) - P(AB) \therefore P(A) = P(A \cup B) - P(B \cap \bar{A}) = 0.5$

(2) $P(A \cap \bar{B}) = P(A) - P(AB) = 0.5 - 0.2 = 0.3$

$P(AB) = P(A) + P(B) - 0.9 = 0.5 + 0.6 - 0.9 = 0.2$

A8  $P(B-A) = 0.1 \quad P(A-B) = 0.2 \quad P(AB) = 0.3$

(1) $P(A\bar{B}) = P(A-B) = 0.2$

(2) $P(\bar{A}\bar{B}) = P(\bar{A}) + P(\bar{B}) - P(\bar{A} \cup \bar{B})$
 $= 0.5 + 0.6 - 1 + P(AB) = 0.4$

(3) $P(\bar{A} \cup \bar{B}) = 1 - P(AB) = 0.7$

A12 事件A: "两次均为红球" 事件B: "恰有一个红球" 事件C: "第二次是红球"

1. 有放回抽样.

(1) $P(A) = \frac{8 \times 8}{10 \times 10} = \frac{16}{25}$

(2) $P(B) = \frac{8 \times 2 \times 2}{10 \times 10} = \frac{8}{25}$

(3) $P(C) = \frac{8 \times 10}{10 \times 10} = \frac{4}{5}$

2. 无放回抽样.

(1) $P(A) = \frac{8 \times 7}{10 \times 9} = \frac{28}{45}$

(2) $P(B) = \frac{C_8^1 C_2^1}{C_{10}^2} = \frac{16}{45}$

(3) $P(C) = \frac{8 \times 9}{9 \times 10} = \frac{4}{5}$

A14 (1) $P("1红1黑1白") = \frac{C_2^1 C_2^1 C_2^1}{C_7^3} = \frac{3 \times 2 \times 2 \times 2 \times 2}{7 \times 6 \times 5} = \frac{12}{35}$

(2) $P("全是黑球") = \frac{C_2^3}{C_7^3} = \frac{1}{35}$

(3) $P("第一次摸红, 第二次摸黑, 第三次摸白") = \frac{2}{7} \times \frac{3}{6} \times \frac{2}{5} = \frac{2}{35}$

B1 (1) $P(C \cap \bar{A}) = P(C-A) = P(C) - P(A) = 0.1$

(2) $P(A \cup B) = P(A) + P(B) - P(AB) = 0.6$

$\therefore A \subset C, B \cap C = \emptyset \therefore P(AB) = 0$

(3) $P(\bar{A} \cap \bar{B} \cap \bar{C}) = 1 - P(A \cup B \cup C) = 1 - P(C) - P(B) = 0.3$

秋二周 A 19. 20. 23. 26. 29. 30. 32. B 6. 7.

$$A19. (1) P(A|A \cup B) = \frac{P(A \cap (A \cup B))}{P(A \cup B)} = \frac{P(A)}{P(A \cup B)} = \frac{7}{9}$$

$$P(A \cup B) = P(A) + P(B) - P(AB) = 0.7 + 0.4 - 0.2 = 0.9$$

$$P(AB) + P(\bar{A}\bar{B}) = P(A) \Rightarrow P(AB) = P(A) - P(\bar{A}\bar{B}) = 0.7 - 0.5 = 0.2$$

$$(2) P(A|\bar{A} \cup B) = \frac{P(A \cap (\bar{A} \cup B))}{P(\bar{A} \cup B)} = \frac{P(AB)}{P(\bar{A}) + P(B) - P(\bar{A}B)} = \frac{0.2}{0.3 + 0.4 - (0.4 - 0.2)} = \frac{2}{5}$$

$$(3) P(AB|A \cup B) = \frac{P(AB \cap (A \cup B))}{P(A \cup B)} = \frac{P(AB)}{P(A \cup B)} = \frac{2}{9}$$



画韦恩图方便判断.

A20. $A_1 = \{\text{选甲盒}\}$ $A_2 = \{\text{选乙盒}\}$ $B = \{\text{取红球}\}$ $B_2 = \{\text{取白球}\}$

(1) $C_1 = \{\text{取一红一白}\}$

$$P(C_1) = \frac{1}{2} \times \frac{C_2^1 C_2^1}{C_5^2} + \frac{1}{2} \times \frac{C_2^2 C_4^1}{C_6^2} = \frac{17}{30}$$

(2) $C_2 = \{\text{从乙盒中取到红球}\}$

$$P(C_2) = \frac{3}{5} \times \frac{3}{7} + \frac{2}{5} \times \frac{2}{7} = \frac{13}{35}$$

(3) $C_3 = \{\text{从乙盒中取到一红一白}\}$

$$P(C_3) = \frac{3}{5} \times \frac{C_2^1 C_4^1}{C_7^2} + \frac{2}{5} \times \frac{C_2^2 C_5^1}{C_7^2} = \frac{56}{105}$$

A23. $A = \{\text{好枪}\}$ $B = \{\text{命中}\}$

$$P(B|A) = \frac{3}{5} \quad P(B|\bar{A}) = \frac{1}{20}$$

$$(1) P(B|A) = \frac{P(AB)}{P(A)} \quad P(B|\bar{A}) = \frac{P(\bar{A}B)}{P(\bar{A})}$$

$$P(B) = P(AB) + P(\bar{A}B) = P(A)P(B|A) + P(\bar{A})P(B|\bar{A}) = \frac{3}{5} \times \frac{3}{4} + \frac{1}{20} \times \frac{1}{4} = \frac{37}{80}$$

$$(2) \text{求 } P(\bar{A}|B) = \frac{P(\bar{A}B)}{P(B)} = \frac{P(\bar{A})P(B|\bar{A})}{P(B)} = \frac{1}{4} \times \frac{1}{20} \times \frac{80}{37} = \frac{1}{37}$$

A26. $A_1 = \{X < 20\}$ $A_2 = \{20 \leq X \leq 40\}$ $A_3 = \{X > 40\}$

$$B = \{\text{获奖}\} \quad P(B|A_1) = 0.6 \quad P(B|A_2) = 0.5 \quad P(B|A_3) = 0.3 \quad P(A_1) = 0.2 \quad P(A_2) = 0.7 \quad P(A_3) = 0.1$$

$$P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3) = 0.5$$

$$P(A_1|B) = P(A_1B) / P(B) = P(A_1)P(B|A_1) / P(B) = 0.24$$

A29. $X = \{\text{元件X正常工作}\}$ $P(A) = P(B) = P(C) = p$

$$P(\text{"系统正常工作"}) = P(C) + P(\bar{C})P(AB|\bar{C}) \\ = p + (1-p)p^2 = p + p^2 - p^3$$

A30 $A_i = \{\text{第}i\text{次命中}\}$

$$(1) P(\text{"前2次都命中"}) = P(A_1)P(A_2|A_1) = 0.64$$

$$(2) P(\text{"至少命中1次"}) = P(A_1) + P(\bar{A}_1)P(A_2|\bar{A}_1) + P(\bar{A}_1\bar{A}_2)P(A_3|\bar{A}_1\bar{A}_2) \\ = 0.8 + 0.2 \times 0.8 + 0.2 \times 0.2 \times 0.8 \\ = 0.992$$

B32 1) $P(A_i) = (1-p)^{i-1} \cdot p, i=1, 2, \dots$

$$P(B_4) = (1-p)^2 p^2 + (1-p)^3 \cdot p = (1-p)^2 \cdot p$$

可能发生的情况要讨论完全

$$(2) P(B_4 | A_1) = \frac{P(A_1 B_4)}{P(A_1)} = \frac{p^3 (1-p)}{p} = p^2 (1-p)$$

$$(3) P(A_1 | B_4) = \frac{P(A_1 B_4)}{P(B_4)} = \frac{p^3 (1-p)}{p^2 (1-p)} = p$$

B6, 第一组: 30件, 10件优 第二组: 20件, 15件优

$A_i = \{ \text{抽到第} i \text{组} \}$ $B_i = \{ \text{第} i \text{次抽到优质品} \}$

$$(1) P(B_1) = P(A_1 B_1) + P(A_2 B_1)$$

$$= \frac{1}{2} \times \frac{1}{3} + \frac{1}{2} \times \frac{3}{4} = \frac{13}{24}$$

$$(2) P(\bar{B}_2 | B_1) = \frac{P(B_1 \bar{B}_2)}{P(B_1)} = 0.394$$

$$P(B_1 \bar{B}_2)$$

$$= \frac{1}{2} \times \frac{10}{30} \times \frac{20}{29} + \frac{1}{2} \times \frac{15}{20} \times \frac{5}{19}$$

$$= \frac{1}{2} \times \frac{1}{3} \times \frac{20}{29} + \frac{1}{2} \times \frac{3}{4} \times \frac{5}{19} = \frac{2825}{7163}$$

B7. 1) $\alpha = p_1 p_2 p_3 + p_1 p_2 p_4 + p_1 p_3 p_4 + p_2 p_3 p_4 - 3 p_1 p_2 p_3 p_4$

$$= p_1 p_2 p_3 p_4 \left(\frac{1}{p_1} + \frac{1}{p_2} + \frac{1}{p_3} + \frac{1}{p_4} - 3 \right)$$

$$(2) \beta = \frac{p_1 p_2 p_3 p_4}{\alpha} = \frac{p_1 p_2 p_3 p_4}{p_1 p_2 p_3 + p_1 p_2 p_4 + p_1 p_3 p_4 + p_2 p_3 p_4 - 2 p_1 p_2 p_3 p_4}$$

$$(3) \chi = \alpha^2 (1-\alpha) \times 3$$

秋三周: 习题 = A 1、6、8、10、12、 B 6、9、16、19

A 1. (1)

X	1	2	3
P	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{6}$

$$S = \{12, 13, 14, 21, 23, 24, 31, 32, 34, 41, 42, 43\}$$

(2)

X	0	1	2	3
P	$\frac{8}{27}$	$\frac{4}{9}$	$\frac{2}{9}$	$\frac{1}{27}$

$$P(\text{"掷一次点数大于4"}) = \frac{1}{3}$$

$$P(X=3) = \left(\frac{1}{3}\right)^3$$

$$P(X=2) = C_3^1 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^1 = \frac{2}{9}$$

$$P(X=1) = C_3^2 \left(\frac{1}{3}\right)^1 \left(\frac{2}{3}\right)^2 = \frac{4}{9}$$

$$P(X=0) = C_3^3 \left(\frac{2}{3}\right)^3 = \frac{8}{27}$$

(3)

X	0	1	2
P	$\frac{3}{10}$	$\frac{3}{5}$	$\frac{1}{10}$

$$P(X=2) = \frac{C_2^2}{C_5^2} = \frac{1}{10}$$

$$P(X=1) = \frac{C_2^1 C_3^1}{C_5^2} = \frac{3}{5}$$

$$P(X=0) = \frac{C_3^2}{C_5^2} = \frac{3}{10}$$

(4)

X	1	2	3
P	$\frac{2}{5}$	$\frac{6}{25}$	$\frac{9}{25}$

$$P(X=1) = \frac{2}{5} \quad P(X=2) = \frac{2}{5} \times \frac{3}{5} \quad P(X=3) = \frac{3}{5} \times \frac{3}{5} \times \frac{2}{5} + \left(\frac{3}{5}\right)^3$$

6. $X \sim P(2) \quad P(X=k) = \frac{2^k e^{-2}}{k!}$

$$P\{X \leq 2\} = P(X=0) + P(X=1) + P(X=2) \\ = e^{-2} + 2e^{-2} + 2e^{-2} = 5e^{-2}$$

$$P\{X \geq 2\} = 1 - \frac{3}{e^2}$$

$$P\{X \leq 1 | X \leq 2\} = \frac{P\{X \leq 1\}}{P\{X \leq 2\}} = \frac{3}{5}$$

8.

$$F(x) = \begin{cases} 0, & x < 0 \\ 0.4, & 0 \leq x < 2 \\ 0.9, & 2 \leq x < 3 \\ 1, & x \geq 3 \end{cases} \quad P\{1 \leq X < 3\} = 0.5$$

10. (1) $\int_{-\infty}^{+\infty} f(x) dx = 1 = \int_0^2 c(2-x) dx = 2cx - \frac{c}{2}x^2 \Big|_0^2 = 4c - 2c = 2c \Rightarrow c = \frac{1}{2}$

(2) $F(x) = \int_{-\infty}^{+\infty} f(x) dx$

$$= \begin{cases} 0, & x < 0 \\ x - \frac{x^2}{4}, & 0 \leq x < 2 \\ 1, & x \geq 2 \end{cases}$$

(3) $P\{0.5 < X \leq 1\} = F(1) - F(0.5)$

$$= 1 - \frac{1}{4} - \frac{1}{2} + \frac{1}{16} = \frac{5}{16}$$

$$12. X \sim U(0,4) \quad f(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{4}, & 0 \leq x < 4 \\ 0, & x \geq 4 \end{cases}$$

$$x^2 - 2x - 3 = (x-3)(x+1) < 0$$

$$\Rightarrow \underline{-1 < x < 3}$$

$$P(-1 < X < 3) = \int_{-1}^3 f(x) dx = \frac{3}{4}$$

$$B6. (1) P(A) = C_5^3 (0.8)^3 (0.2)^2 = 0.2048$$

$$(2) P(B) = C_5^4 (0.8)^4 (0.2)^1 + (0.8)^5 = 0.73728$$

$$(3) P(C) = (0.2)^5 + C_5^1 (0.8)^1 (0.2)^4 + C_5^2 (0.8)^2 (0.2)^3 = 0.05792$$

$$9. X \sim P(1)$$

$$F(x) = \frac{1^k e^{-1}}{k!} = \frac{1}{e \cdot k!}$$

$$(1) P\{X \geq 2\} = 1 - P(X=0) - P(X=1) = 1 - \frac{2}{e}$$

$$(2) P\{X \leq 3 | X \geq 2\} = \frac{P(X=2) + P(X=3)}{P\{X \geq 2\}} = \frac{\frac{1}{2e} + \frac{1}{6e}}{1 - \frac{2}{e}} = \frac{2}{3e-6}$$

$$16. (1) F(x) = \begin{cases} 0, & x < 0 \\ \frac{x}{2}, & 0 \leq x < 1 \\ \frac{1}{2}, & 1 \leq x < 2 \\ \frac{x-1}{2}, & 2 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$

$$(2) P\{X \leq 2.5\} = F\left(\frac{5}{2}\right) = \frac{3}{4}$$

$$19. (1) P(A) = \frac{30-20+45-40}{45-20} = \frac{3}{5}$$

$$(2) P(B) = \frac{25-20+45-35}{45-20} = \frac{3}{5}$$

$$(3) C: \text{等车时间} \geq 75 \text{ min}$$

$$D: \text{赶上} 7:30.$$

$$P(C) = \frac{25-20+45-30}{45-20} = \frac{4}{5}$$

$$P(D) = \frac{30-20}{45-20} = \frac{2}{5}$$

$$P(CD) = \frac{25-20}{45-20} = \frac{1}{5}$$

$$\therefore P(C|C) = \frac{P(CD)}{P(C)} = \frac{1}{4}$$

概率论第4次作业: 习题 A 12, 14 B 18, 22, 25, 27, 33, 36

$$12. X \sim U(0, 4) \quad f(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{4}, & 0 \leq x < 4 \\ 0, & x \geq 4 \end{cases}$$

$$x^2 - 2x - 3 = (x-3)(x+1) < 0$$

$$\Rightarrow \underline{-1 < x < 3}$$

$$P(-1 < X < 3) = \int_{-1}^3 f(x) dx = \frac{3}{4}$$

$$14. (1) F(x) = \begin{cases} 1 - e^{-0.2x} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

$$(2) P\{X > 5\} = 1 - F(5) = \frac{1}{e}$$

$$(3) P\{5 < X \leq 10\} = F(10) - F(5) = \frac{1}{e} - \frac{1}{e^2}$$

$$P\{X \leq 10 | X > 5\} = \frac{P\{5 < X \leq 10\}}{P(X > 5)} = 1 - \frac{1}{e}$$

B 18.

$$F(x) = \begin{cases} 0, & x < 0 \\ ax^2, & 0 \leq x < 1 \\ bx, & 1 \leq x < 2 \\ 1, & 2 \leq x \end{cases}$$

$$(1). f(x) = \begin{cases} 2ax, & 0 \leq x < 1 \\ b, & 1 \leq x < 2 \\ 0, & \text{其他.} \end{cases}$$

$$\int_{-\infty}^{+\infty} f(x) dx = 1 = \int_0^1 2ax dx + \int_1^2 b dx = 1 = a + b$$

x 为连续型随机变量: 故 $F(2) = F(2-0) = 1 = 2b \Rightarrow a = b = \frac{1}{2}$

$$(2) f(x) = \begin{cases} 2ax, & 0 \leq x < 1 \\ b, & 1 \leq x < 2 \\ 0, & \text{其他.} \end{cases}$$

$$(3) P\{0.5 < X < 1.5\} = F(1.5) - F(0.5) - P(1.5) = \frac{1.5}{2} - \frac{1}{4} \times \frac{1}{2} - 0 = \frac{5}{8}$$

$$22. X \sim N(170, 5.0^2)$$

$$(1) P(X > 170) = 1 - \Phi\left(\frac{170-170}{5}\right) = \frac{1}{2}$$

$$(2) P(165 < X < 175) = 2\Phi\left(\frac{175-170}{5}\right) - 1 \approx 0.6826$$

$$(3) P(X < 172) = \Phi\left(\frac{172-170}{5}\right) = \Phi\left(\frac{2}{5}\right) \approx 0.6554$$

$$25. P(X < 12800) = 30\%$$

$$P(X > 10000) = 95\%$$

$$\begin{cases} 0.3 = \Phi\left(\frac{\mu - 12800}{\sigma}\right) \\ 0.95 = \Phi\left(\frac{10000 - \mu}{\sigma}\right) \end{cases} \Rightarrow \begin{cases} \left|\frac{\mu - 12800}{\sigma}\right| \approx 0.5240 \\ \left|\frac{10000 - \mu}{\sigma}\right| \approx 1.6449 \end{cases} \Rightarrow \begin{cases} \mu = 14109 \\ \sigma = 2498 \end{cases}$$

27. (1) $\int_{-\infty}^{+\infty} f(x) dx = 1 = \int_{-\infty}^{+\infty} a \cdot e^{-x^2} dx = a \cdot \sqrt{\pi} \Rightarrow a = \frac{1}{\sqrt{\pi}}$

(2) $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

$f(x) = \frac{1}{\sqrt{\pi}} e^{-x^2} \Rightarrow X \sim N(0, (\frac{1}{\sqrt{2}})^2)$

$P\{X > \frac{1}{2}\} = \Phi\left(\frac{\frac{1}{2}-0}{\frac{1}{\sqrt{2}}}\right) = \Phi\left(\frac{\sqrt{2}}{2}\right) =$

33 $f(x) = \begin{cases} c(4-x^2) & -1 < x < 2 \\ 0 & \text{其他} \end{cases}$

(1) $\int_{-\infty}^{+\infty} f(x) dx = 1 = \int_{-1}^2 c(4-x^2) dx = 12c - \frac{c}{3} x^3 \Big|_{-1}^2 = 9c \Rightarrow c = \frac{1}{9}$

(2) $Y = 3X \quad X = \frac{Y}{3}$

$f(y) = \begin{cases} \frac{1}{27} (4 - (\frac{Y}{3})^2) & -3 < y < 6 \\ 0 & \text{其他} \end{cases}$

(3) $f(z) = \begin{cases} \frac{2}{9} (4-z^2), & 0 < z < 1 \\ \frac{1}{9} (4-z^2), & 1 < z < 2 \\ 0, & \text{其他} \end{cases}$

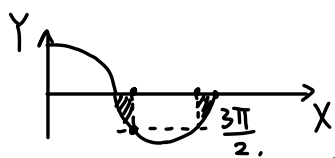
$F(z) = \begin{cases} 0, & z < 0 \\ \frac{2}{9} (4z - \frac{z^3}{3}), & 0 \leq z < 1 \\ \frac{1}{9} (4z - \frac{z^3}{3}) + \frac{11}{27}, & 1 \leq z < 2 \\ 1, & z \geq 2 \end{cases}$

②

36. $X \sim U(0, \frac{3\pi}{2})$

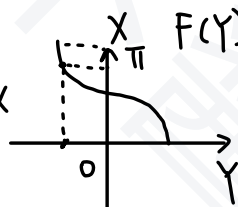
$f(x) = \begin{cases} \frac{2}{3\pi}, & 0 < x < \frac{3\pi}{2} \\ 0, & \text{其他} \end{cases}$

$F(x) = \begin{cases} 0, & x < 0 \\ \frac{2}{3\pi} x, & 0 \leq x < \frac{3\pi}{2} \\ 1, & x \geq \frac{3\pi}{2} \end{cases}$



$X = \arccos Y$

$F(y) = P(Y \leq y) = P(\cos X \leq y)$



$F(y) = \begin{cases} 0, & y < -1 \\ \frac{4}{3\pi} (\pi - \arccos y) \\ 1 - \frac{2}{3\pi} \arccos y \\ 1, & y \geq 1 \end{cases}$

$-1 \leq y < 0$
 $0 \leq y < 1$

概率论第五次. 张婧怡 3230100952

习题三: A2. A8. A11. A16. A19. A21. B4. B5.

A2. (1): $0.6 = 0.1x + a + 2a$, $a = 0.1$ $b = 1 - 0.6 - 0.2 = 0.2$

(2): $P\{X=0|Y=0\} = 0.5 = \frac{0.1}{0.1+0+b} \Rightarrow b=0.1$ $1-0.1-0.2 = 0.1x+3a \Rightarrow a = \frac{4}{30} = \frac{2}{15}$

(3): $P\{X \leq 1, Y \geq 1\} = 3a + 0.2 = 0.35 \Rightarrow a = 0.05$ $b + 0.2 + 0.1 + 0.35 = 1 \Rightarrow b = 0.35$

A8 (1) (X, Y) 的联合分布律.

X	Y		
	0	1	2
0	0.2	0.1	0.1
1	0	0.4	0.2

(2) $P\{Y=0|X=0\} = \frac{p_{ij}}{p_{i.}} = 0.5$ $P\{Y=1|X=0\} = 0.25$ $P\{Y=2|X=0\} = 0.25$

A11. (1). $F(0, 1) = P\{X \leq 0, Y \leq 1\} = 0.2$

$F(1, 1.5) = P\{X \leq 1, Y \leq 1.5\} = 0.4$

$F(2, 1, 1) = P\{X \leq 2, Y \leq 1\} = 0.6$

(2). $F_X(x) = P\{X \leq x, Y \leq +\infty\}$

$$= \begin{cases} 0 & x < 0 \\ 0.2 & 0 \leq x < 1 \\ 0.6 & 1 \leq x < 2 \\ 1 & x \geq 2 \end{cases}$$

A16. (1). $\int_1^2 \int_x^{4-x} c(x-1) dy dx = \int_1^2 c(x-1)(4-2x) dx = c(1 - \frac{2}{3}x^3 + 3x^2 - 4x) \Big|_1^2 = \frac{c}{3} = 1 \Rightarrow C=3$

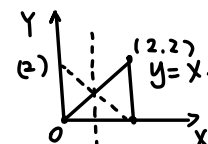
(2) $f_X(x) = \begin{cases} 6(x-1)(2-x), & 1 < x < 2 \\ 0, & \text{其他.} \end{cases}$ $f_Y(y) = \begin{cases} \frac{3(y-1)^2}{2}, & 1 < y \leq 2 \\ \frac{3(3-y)^2}{2}, & 2 < y < 3. \\ 0, & \text{其他.} \end{cases}$

A19 $f_X(x) = \begin{cases} \lambda^2 x e^{-\lambda x}, & x > 0 \\ 0, & x \leq 0. \end{cases}$ ($\lambda > 0$) $f_{Y|X}(y|x) = \begin{cases} \frac{1}{x} e^{-y/x}, & y > 0 \\ 0, & y \leq 0 \end{cases}$

(1). $f(x, y) = f_X(x) \cdot f_{Y|X}(y|x) = \begin{cases} \lambda^2 e^{-\lambda x} e^{-y/x}, & x > 0 \text{ 且 } y > 0 \\ 0, & x \leq 0, y \leq 0 \end{cases}$

(2) $\int f_{Y|X}(y|x) dy = -e^{-\frac{y}{x}} + C$. ($y > 0$) $y \rightarrow +\infty, F_{Y|X}(y|x) \rightarrow 1$
 $\therefore F_{Y|X}(y|x) = \begin{cases} 1 - e^{-\frac{y}{x}}, & y > 0 \\ 0, & y \leq 0 \end{cases}$

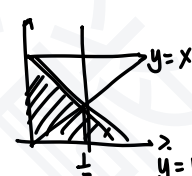
(3) $P\{Y > 1|X=1\} = 1 - P\{Y \leq 1|X=1\} = 1 - (1 - e^{-1}) = e^{-1}$

A21. (2)  (1) $f(x, y) = \begin{cases} \frac{1}{2} & 0 \leq y \leq x \leq 2 \\ 0 & \text{其他} \end{cases}$

(2) $P\{X+Y \geq 2\} = \frac{1}{2} = 0.5$

(3) $P\{X < 1\} = \frac{1}{4} = 0.25$

B4. (1) $\int_{-\infty}^{\infty} \int_{-\infty}^y f(x, y) dx dy = 1. = \int_0^1 \int_x^1 c(y-x) dy dx = \int_0^1 (\frac{cx^2}{2} - cx + \frac{c}{2}) dx = c \frac{x^3}{6} \Big|_0^1 = \frac{c}{6} = 1 \Rightarrow c=6$

(2)  $0 < x < y < 1. \quad x+y \leq 1.$
 $\therefore P\{X+Y \leq 1\} = \int_0^{\frac{1}{2}} \int_x^{1-x} 6(y-x) dy dx$
 $= \int_0^{\frac{1}{2}} (3y^2 - 6xy) \Big|_x^{1-x} dx = \int_0^{\frac{1}{2}} (-12x^2 + 12x - 3) dx = 4x^3 - 6x^2 + 3x \Big|_0^{\frac{1}{2}} = \frac{1}{8}$

(3). $P\{X < \frac{1}{2}\} = \int_0^{\frac{1}{2}} \int_x^1 6(y-x) dy dx = \int_0^{\frac{1}{2}} (3y^2 - 6xy) \Big|_x^1 dx = \int_0^{\frac{1}{2}} (3x^2 - 6x + 3) dx = x^3 - 3x^2 + 3x \Big|_0^{\frac{1}{2}} = \frac{7}{8}$

B5 (1). $f_X(x) = \begin{cases} 1, & 1 < x < 2 \\ 0, & \text{其他} \end{cases}$

$f(x, y) = f_X(x) \cdot f_{Y|X}(y|x) = \begin{cases} \frac{2(4-y)}{(3-x)^2} & 1 < x < 2, \quad x+1 < y < 4 \\ 0, & \text{其他} \end{cases}$

$P\{Y < 3\} = \int_1^2 \int_{x+1}^3 \frac{2(4-y)}{(3-x)^2} dy dx = \int_1^2 \frac{8(2-x)}{(3-x)^2} - \frac{9-(1+x)^2}{(3-x)^2} dx = \int_1^2 \frac{x^2 - 6x + 8}{(3-x)^2} dx$
 $= 1 - (1 - \frac{1}{2}) = \frac{1}{2}$

(2) $f_Y(y) = \begin{cases} y-2 & 2 < y < 3 \\ 4-y & 3 < y < 4 \\ 0 & \text{其他} \end{cases}$

(3). $Y=3 \quad P\{X \leq 1.5 | Y=3\} = \int_1^{1.5} \frac{2}{(3-x)^2} dx = \frac{2}{3-x} \Big|_1^{1.5} = 2(\frac{1}{1.5} - \frac{1}{2}) = \frac{1}{3}$

概率论第六次作业. 张婧怡 20100952.

习题: A23 A26(3) A32 A36 B10 B11

A23. (1) $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad -\infty < x < +\infty$
 $f_Y(y) = \frac{1}{2\sqrt{\pi}} e^{-\frac{(y-1)^2}{4}} \quad -\infty < y < +\infty$

(2) 在 $\{X=x\}$ 条件下, Y 的条件分布即 $Y \sim N(\mu_2 + \rho \frac{\sigma_2}{\sigma_1}(x - \mu_1), (1-\rho^2)\sigma_2^2)$

$$f_{Y|X}\{y|X=0\} = \frac{1}{\sqrt{2\pi} \sigma_2 \sqrt{1-\rho^2}} \cdot \exp \left\{ -\frac{[y - (\mu_2 + \rho \frac{\sigma_2}{\sigma_1}(0 - \mu_1))]^2}{2 \cdot (1-\rho^2) \cdot \sigma_2^2} \right\}$$

$$= \frac{1}{\sqrt{3\pi}} \cdot e^{-\frac{(y-1)^2}{3}} \quad -\infty < y < +\infty$$

(3) $P\{Y \leq 1 | X=0\} = \int_{-\infty}^1 f_{Y|X}\{y|X=0\} dy = \int_{-\infty}^1 \frac{1}{\sqrt{3\pi}} e^{-\frac{(y-1)^2}{3}} dy = \frac{1}{2}$

A26(3) $f_X(x) = \int_{-\infty}^{+\infty} f(x,y) dy$
 $= \int_0^2 \frac{3x}{4} dy = \frac{3x}{4}(2-y)$

$f_Y(y) = \int_{-\infty}^{+\infty} f(x,y) dx$
 $= \int_0^y \frac{3x}{4} dx = \frac{3}{8} y^2$

$f(x,y) \neq f_X(x) \cdot f_Y(y) \quad \therefore X, Y$ 不独立.

A32

Z	1	2	3	4	5
P	0.04	0.14	0.3	0.32	0.2

M	1	2	3
P	0.1	0.5	0.4

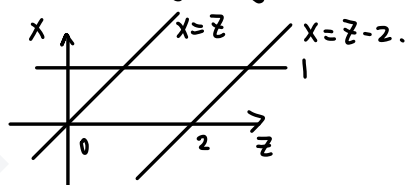
N	0	1	2
P	0.2	0.4	0.4

A36. $Y = 2X - Z \quad 0 < 2X - Z < 2X \quad f(x,z) = \begin{cases} \frac{1}{4}, & 0 < z < 4, \frac{z}{2} < x < z \\ 0, & \text{其他.} \end{cases}$
 $\Rightarrow 0 < z < 4, \quad \frac{z}{2} < x < z$

$\therefore \int_{-\infty}^{+\infty} f(x,z) dx = \int_{\frac{z}{2}}^z \frac{1}{4} dx = \frac{1}{4} (2 - \frac{z}{2}) = \frac{1}{2} - \frac{z}{8} \quad (0 < z < 4)$

$\therefore f_Z(z) = \begin{cases} \frac{1}{2} - \frac{z}{8}, & 0 < z < 4 \\ 0, & \text{其他.} \end{cases}$

B10. $Z = X + Y$, $Y = Z - X$ $0 < Z - X < 2$. $X < Z < 2 + X$



① $0 < Z < 1$. $\int_0^Z \frac{3-z}{3} dx = \frac{(3-z)z}{3}$

② $1 < Z < 2$ $\int_0^1 \frac{3-z}{3} dx = \frac{3-z}{3}$

③ $2 < Z < 3 = \int_2^1 \frac{3-z}{3} dx = \frac{(3-z)(1-z)}{3}$

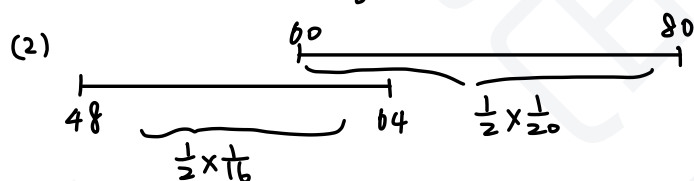
$$\therefore f_Z(z) = \begin{cases} \frac{(3-z)z}{3}, & 0 < z < 1 \\ \frac{3-z}{3}, & 1 < z < 2 \\ \frac{(3-z)(1-z)}{3}, & 2 < z < 3 \\ 0, & \text{其他.} \end{cases}$$

B11. $f_X(x) = \begin{cases} \frac{1}{2}, & 6 < x < 8 \\ 0, & \text{其他.} \end{cases}$

Y	8	10
P	0.5	0.5

$Z = XY$.

(1) $P = P(Z \leq 60) = 0.5 \times (\frac{60}{8} - 6) \times 0.5 + 0.5 \times (\frac{60}{10} - 6) \times 0.5$
 $= \frac{3}{8}$



$$F_Z(z) = \begin{cases} 0, & z < 48 \\ \frac{(z-48)}{32}, & 48 \leq z < 60 \\ \frac{3}{8} + \frac{9}{160}(z-60), & 60 \leq z < 64 \\ \frac{3}{5} + \frac{1}{40}(z-64), & 64 \leq z < 80 \\ 1, & z \geq 80 \end{cases} = \begin{cases} 0 & z < 48 \\ \frac{z}{32} - \frac{3}{2} & 48 \leq z < 60 \\ \frac{9z}{160} - 3 & 60 \leq z < 64 \\ \frac{z}{40} - 1 & 64 \leq z < 80 \\ 1 & z \geq 80 \end{cases}$$

下次作业.

秋周 习题四: A 1, 4, 5, 10, 11 B. 1, 6, 15

A1. (1) 无放回

X	1	2	3
P	$\frac{3}{10}$	$\frac{3}{5}$	$\frac{1}{10}$

$$E(X) = \sum_{i=1}^3 p_i x_i = \frac{3}{10} \times 1 + \frac{3}{5} \times 2 + \frac{1}{10} \times 3 = \frac{9}{5}$$

(2) 有放回

X	0	1	2	3
P	$\frac{8}{125}$	$\frac{36}{125}$	$\frac{54}{125}$	$\frac{27}{125}$

$$E(X) = \sum_{i=1}^4 p_i x_i = \frac{36}{125} \times 1 + \frac{54}{125} \times 2 + \frac{27}{125} \times 3 = \frac{225}{125} = \frac{9}{5}$$

A4. (1)

Z	0	1	2	4
P	0.4	0.1	0.3	0.2

$$E(Z) = \sum_{i=1}^4 z_i p_i = 1.5$$

(2)

Z	0	1	2
P	0.4	0.4	0.2

$$E(Z) = \sum_{i=1}^3 z_i p_i = 0.8$$

(3)

Z	1	2
P	0.3	0.7

$$E(Z) = 0.3 + 2 \times 0.7 = 1.7$$

A5

$$f_X(x) = \int_0^2 \frac{2x+y}{4} dy = x + \frac{y^2}{8} \Big|_0^2 = x + \frac{1}{2}$$

$$\therefore E(X) = \int_0^1 x \cdot f_X(x) dx = \int_0^1 (x^2 + \frac{x}{2}) dx = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$$

$$f_Y(y) = \int_0^1 \frac{2x+y}{4} dx = \frac{1}{4} + \frac{y}{4}$$

$$\therefore E(Y) = \int_0^2 y \cdot (\frac{1}{4} + \frac{y}{4}) dy = \frac{4}{8} + \frac{8}{12} = \frac{1}{2} + \frac{2}{3} = \frac{7}{6}$$

$$\begin{aligned} E(XY) &= \int_0^1 \int_0^2 xy \left(\frac{2x+y}{4} \right) dy dx \\ &= \int_0^1 \int_0^2 \left(\frac{x^2 y}{2} + \frac{xy^2}{4} \right) dy dx \\ &= \int_0^1 \left(x^2 + \frac{2}{3}x \right) dx = \frac{1}{3} + \frac{1}{3} = \frac{2}{3} \end{aligned}$$

A10 (1)

X	0	1	2
P	0.3	0.2	0.5

$$E(X) = \sum_{i=1}^3 p_i x_i = 1.2$$

$$\begin{aligned} \text{Var}(X) &= E\{[X - E(X)]^2\} \\ &= 0.3 \times 1.2^2 + 0.2 \times 0.2^2 + 0.5 \times 0.8^2 \\ &= 0.76 \end{aligned}$$

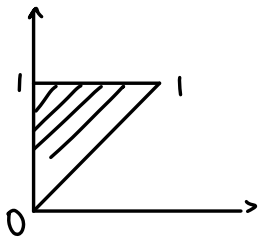
(2)

$$f(x) = \begin{cases} x, & 0 < x < 1 \\ \frac{1}{2}, & 1 < x < 2 \\ 0, & \text{其他} \end{cases}$$

$$\begin{aligned} E(X) &= \int x \cdot f(x) dx = \int_0^1 x^2 dx + \int_1^2 \frac{x}{2} dx \\ &= \frac{1}{3} + \frac{3}{4} = \frac{13}{12} \end{aligned}$$

$$\begin{aligned} \text{Var}(X) &= E(X^2) - [E(X)]^2 \\ &= \int x^2 f(x) dx - \left(\frac{13}{12} \right)^2 = \frac{35}{144} \end{aligned}$$

A11



$$\begin{aligned}\text{Var}(XY) &= E\{[XY - E(XY)]^2\} \\ &= E(X^2Y^2) - [E(XY)]^2 \\ f(x, y) &= \begin{cases} 2, & 0 \leq x \leq y \leq 1 \\ 0, & \text{其他.} \end{cases}\end{aligned}$$

$$E(XY) = \int_0^1 \int_0^x 2xy \, dx \, dy = \int_0^1 x^3 \, dx = \frac{1}{4}$$

$$E(X^2Y^2) = \int_0^1 \int_0^x 2x^2y^2 \, dx \, dy = \int_0^1 \frac{2}{3}x^5 \, dx = \frac{2}{3} \times \frac{1}{6} = \frac{1}{9}$$

$$\therefore \text{Var}(XY) = \frac{1}{9} - \frac{1}{16} = \frac{7}{144}$$

B1. 抽取 n 次. 抽到 X 件正品

$$P(X=k) = \left(\frac{N}{M}\right)^k \cdot \left(\frac{M-N}{M}\right)^{n-k}$$

$$E(X) = \sum_{k=1}^n k \cdot \left(\frac{N}{M}\right)^k \cdot \left(\frac{M-N}{M}\right)^{n-k}$$

$$= \frac{nN}{M}$$

★ N 件正品. M 件商品.每次抽取到正品的概率为 $\frac{N}{M}$. 符合 n 重伯努利试验

$$\text{抽 } n \text{ 次. } \Rightarrow E(X) = \frac{nN}{M}$$

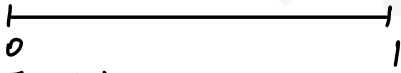
$$B6. (1) f_X(x) = \int_0^x \frac{2}{x} e^{-2x} \, dy = 2e^{-2x}$$

$$\begin{aligned}E(X) &= \int_0^{+\infty} x f_X(x) \, dx = \int_0^{+\infty} 2xe^{-2x} \, dx = \int_0^{+\infty} -x \, de^{-2x} \\ &= -xe^{-2x} \Big|_0^{+\infty} - \int_0^{+\infty} e^{-2x} \, dx \\ &= \int_0^{+\infty} e^{-2x} \, dx = \frac{1}{2}\end{aligned}$$

$$(2) E(3X-1) = 3E(X) - 1 = \frac{1}{2}$$

$$\begin{aligned}(3) E(XY) &= \int_0^{+\infty} \int_0^x xy \cdot \frac{2}{x} e^{-2x} \, dy \, dx \\ &= \int_0^{+\infty} y^2 e^{-2x} \Big|_0^x \, dx \\ &= \int_0^{+\infty} x^2 e^{-2x} \, dx \\ &= -\frac{1}{2} \int_0^{+\infty} x^2 \, de^{-2x} \\ &= -\frac{1}{2} x^2 e^{-2x} - \left(-\frac{1}{2}\right) \int_0^{+\infty} e^{-2x} \, dx^2 \\ &= \int_0^{+\infty} x e^{-2x} \, dx \\ &= -\frac{1}{2} \int_0^{+\infty} x \, de^{-2x} \\ &= +\frac{1}{2} \int_0^{+\infty} e^{-2x} \, dx = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}\end{aligned}$$

B15



取 n 个点.

设取到的第 i 个点为 X_i

$$X_i \sim U(0, 1)$$

n 个点把线段切割为 $n+1$ 段. 长度为 d_i

$$\sum_{i=1}^{n+1} d_i = 1.$$

由于随机分布, 故 $E d_i = \frac{1}{n+1}$.

$$\therefore \text{最远距离 } E d = (n+1) E d_i = \frac{n+1}{n+1}$$

概率论第八次作业 A: 16, 17, 19 B: 20, 24, 29

16.

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

$$E(XY) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy f(x, y) dx dy = \int_0^2 \int_x^2 \frac{3xy}{4} dy dx = \int_0^2 \frac{3x^2}{8} (4-y^2) dx = \frac{8}{5}$$

$$E(X) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x f(x, y) dx dy = \int_0^2 \int_x^2 \frac{3x^2}{4} dy dx = 1$$

$$E(Y) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} y f(x, y) dx dy = \int_0^2 \int_x^2 \frac{3xy}{4} dy dx = \frac{3}{2}$$

$$\therefore \text{Cov}(X, Y) = 0.1$$

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

$$E(X^2) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x^2 f(x, y) dx dy = 1.2$$

$$E(Y^2) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} y^2 f(x, y) dx dy = 0.4$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 1.2 - 1 = 0.2 \quad \text{Var}(Y) = E(Y^2) - [E(Y)]^2 = 0.4 - 0.25 = 0.15$$

$$\therefore \rho_{XY} = \frac{-0.1}{\sqrt{0.2 \times 0.15}} = -\frac{\sqrt{5}}{3}$$

17. $E(X) = -1, E(Y) = 1, \text{Var}(X) = 4, \text{Var}(Y) = 4, \rho = 0.6$

$$\rho \cdot \sqrt{\text{Var}(X)\text{Var}(Y)} = \text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

$$\therefore E(XY) = 0.6 \times 4 + (-1) \times 1 = 1.4$$

19. $(X+Y)$ 与 $(X-aY)$ 相互独立, 则相关系数为 0 $\text{Cov}(X, Y) = \rho \cdot \sqrt{\text{Var}(X)\text{Var}(Y)} = 2.4$

$$\text{Cov}(X+Y, X-aY) = E[(X+Y)(X-aY)] - E(X+Y)E(X-aY)$$

$$\begin{aligned} E[(X+Y)(X-aY)] &= E[X^2 + (1-a)XY - aY^2] = E(X^2) + (1-a)E(XY) - aE(Y^2) \\ &= \text{Var}(X) + E^2(X) + (1-a)[\text{Cov}(X, Y) + E(X)E(Y)] - a[\text{Var}(Y) + E^2(Y)] \\ &= 4 + 1 + (1-a)(2.4 - 1) - a(4 + 1) = 6.4(1-a) \end{aligned}$$

$$E(X+Y) = E(X) + E(Y) = 0$$

$$E(X-aY) = E(X) - aE(Y) = -1 - a$$

$$6.4(1-a) = 0 \Rightarrow a = 1$$

20. $\text{Var}(X) = E(X^2) - [E(X)]^2$

$$\begin{aligned} E(X) &= \int_{-\infty}^{+\infty} x f(x) dx = \int_0^{+\infty} x \cdot \frac{1}{2} e^{-x} dx + \int_{-\infty}^0 x \cdot \frac{1}{2} e^x dx \\ &= \int_0^{+\infty} -\frac{x}{2} d e^{-x} + \int_{-\infty}^0 \frac{x}{2} d e^x = -\frac{x}{2} e^{-x} \Big|_0^{+\infty} - \int_0^{+\infty} e^{-x} d \frac{x}{2} + \frac{x}{2} e^x \Big|_{-\infty}^0 - \int_{-\infty}^0 e^x d \frac{x}{2} \\ &= -\frac{1}{2} e^{-x} \Big|_0^{+\infty} - \frac{1}{2} e^x \Big|_{-\infty}^0 = \frac{1}{2} - \frac{1}{2} = 0 \end{aligned}$$

$$\begin{aligned} E(X^2) &= \int_{-\infty}^{+\infty} x^2 f(x) dx = \int_0^{+\infty} \frac{x^2}{2} e^{-x} dx + \int_{-\infty}^0 \frac{x^2}{2} e^x dx \\ &= \int_0^{+\infty} x e^{-x} dx - \int_{-\infty}^0 x e^x dx \\ &= -e^{-x} \Big|_0^{+\infty} + e^x \Big|_{-\infty}^0 = 2 \end{aligned}$$

$$\text{Var}(X) = 2$$

$$\text{Var}(X|Y) = E(X^2|Y) - [E(X|Y)]^2 = 2 - [E(X|Y)]^2$$

$$E(X|Y) = 2 \times \int_0^\infty x \frac{1}{2} e^{-x} dx = 1$$

$$\therefore \text{Var}(X|Y) = 1.$$

$$24. \rho = \frac{\text{Cov}(X, |X|)}{\sqrt{\text{Var}(X) \text{Var}(|X|)}}$$

$$\text{Cov}(X, |X|) = E(X|X|) - E(X)E(|X|)$$

$$E(X|X|) = \int_0^\infty \frac{x^2}{2} e^{-x} dx + \int_{-\infty}^0 -\frac{x^2}{2} e^{-x} dx = 0$$

$$\therefore \text{Cov}(X, |X|) = 0. \quad \rho = 0. \quad X \text{ 与 } |X| \text{ 不相关}$$

$$25. P(0 \leq X \leq 1, 0 \leq |X| \leq 1) = P(0 \leq X \leq 1) = \int_0^1 \frac{1}{2} e^{-x} dx = \frac{1}{2}(1 - e^{-1})$$

$$P(0 \leq X \leq 1) \cdot P(0 \leq |X| \leq 1) = \left(\frac{1 - e^{-1}}{2}\right)^2 \neq P(0 \leq X \leq 1, 0 \leq |X| \leq 1) \quad \therefore X, |X| \text{ 不独立}$$

$$29. f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad \varphi(\varepsilon) \text{ 为 } X \sim N(0, 1) \text{ 分布函数}, E(Y) = p - (1-p) = 2p-1$$

$$X, Y \text{ 相互独立}, \text{Cov}(X, Y) = 0$$

$$E(\xi) = E(XY) = \text{Cov}(X, Y) + E(X)E(Y) = 0$$

$$\text{Var}(\xi) = \text{Var}(XY) = E[(XY - E(XY))]^2 = E[X^2 Y^2] = E[X^2, Y^2] = 1$$

$$F_\xi(\varepsilon) = P\{\xi \leq \varepsilon\} = P\{XY \leq \varepsilon\} = P\{XY \leq \varepsilon, Y=1\} + P\{XY \leq \varepsilon, Y=-1\}$$

$$= P\{Y=0\} + P\{XY \leq \varepsilon, Y=1\} + P\{Y=-1\} + P\{XY \leq \varepsilon, Y=-1\}$$

$$= p P\{X \leq \varepsilon | Y=1\} + (1-p) P\{-X \leq \varepsilon, Y=-1\}$$

$$= p \varphi(\varepsilon) + (1-p)(1 - \varphi(-\varepsilon))$$

$$= \varphi(\varepsilon)$$

$$\therefore \xi \sim N(0, 1)$$

$$(2) \rho_{X\xi} = \frac{\text{Cov}(X, \xi)}{\sqrt{\text{Var}(X) \text{Var}(\xi)}} = E(X\xi) - E(X)E(\xi) = E(X^2 Y)$$

$$= p E(X^2) + (1-p) E(-X^2) = 2p-1.$$

当 $p = \frac{1}{2}$, X 与 ξ 不相关. $p > \frac{1}{2}$, X 与 ξ 正相关. $p < \frac{1}{2}$, X 与 ξ 负相关.

$$X=0 \text{ 时}, P(X \leq 0) = 0.5 \quad P(X\xi \leq 0) = P(X^2 Y \leq 0) = p \cdot P(X^2 \leq 0) = 0$$

$$P(\xi \leq 0) = P(XY \leq 0) = 0.5p + 0.5(1-p) = 0.5$$

$\therefore X, \xi$ 在 $0 < p < 1$ 时不独立.

随机试验 $E \rightarrow$ 样本空间 $S = \{e\}$. 样本点 e .

“基本事件”：一个元素的单点集.

样本空间的子集不一定都是随机事件.

- $P(A) = 1 - P(\bar{A})$

- $P(B - A) = P(B - AB) = P(B) - P(AB)$

- $P(A \cup B) = P(A) + P(B) - P(AB)$

$$\Rightarrow P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i) - \sum_{1 \leq i < j \leq n} P(A_i A_j) + \sum_{1 \leq i < j < k \leq n} P(A_i A_j A_k) \cdots + (-1)^{n-1} P(A_1 A_2 \cdots A_n)$$

古典概型：每一样本点等可能： $P(A) = \frac{\text{事件} A \text{ 样本点数}}{\text{样本空间} S \text{ 样本点数}}$

事件互不相容 = 互斥： $A \cap B = \emptyset$

和、交与逆事件具有以下的运算规则：

交换律： $A \cup B = B \cup A, A \cap B = B \cap A$;

结合律： $A \cup (B \cap C) = (A \cup B) \cap C, A(B \cap C) = (AB)C$;

分配律： $A(B \cup C) = (AB) \cup (AC), (AB) \cup C = (A \cup C)(B \cup C)$;

德摩根定律 (De Morgan's law):

$$\overline{\bigcup_{j=1}^n A_j} = \bigcap_{j=1}^n \bar{A}_j, \quad \overline{\bigcap_{j=1}^n A_j} = \bigcup_{j=1}^n \bar{A}_j,$$

★ 设定好基本的样本点..用基础的样本点去表示所求概率事件..再活用公式.

→ 缩减样本空间.

条件概率: $\because P(B) > 0$, 在 B 发生的条件下 A 发生的条件概率. $P(A|B) = \frac{P(AB)}{P(B)}$

性质: $P(C) \neq 0$:

1. $P(A|C) \geq 0$
2. $P(S|C) = 1$
3. $P(B|C) = 1 - P(\bar{B}|C)$
4. $A \supset B$ 时, $P(A|C) \geq P(B|C)$
5. $P(A \cup B|C) = P(A|C) + P(B|C) - P(AB|C)$
 $\hookrightarrow$ 若 $AB = \emptyset \Rightarrow P(A \cup B|C) = P(A|C) + P(B|C)$

条件概率乘法公式: $P(AB) = P(A)P(B|A) = P(B) \cdot P(A|B)$

$$P(A_1 A_2 \cdots A_n) = P(A_1) P(A_2|A_1) P(A_3|A_1 A_2) \cdots$$

$$P(AB|C) = P(A|C) P(B|AC)$$

假设 A, B 为随机事件, 那么总是有 $A = AB \cup A\bar{B}$, 又因为 AB 与 $A\bar{B}$ 不相容, 可得

$$P(A) = P(AB) + P(A\bar{B}) = P(B) \cdot P(A|B) + P(\bar{B}) \cdot P(A|\bar{B}). \quad (1.4.5)$$

定义 1.4.2 设 S 为某一随机试验的样本空间, $B_1, B_2, \dots, B_n$ 为该试验的一组事件, 且满足

$$(1) B_i B_j = \emptyset, i, j = 1, 2, \dots, n, i \neq j;$$

$$(2) B_1 \cup B_2 \cup \cdots \cup B_n = S,$$

则称 $B_1, B_2, \dots, B_n$ 为 S 的一个划分, 或称为 S 的一个完备事件组.

定理 1.4.1 设 S 为某一试验的样本空间, A 为该试验的事件. 设 $B_1, B_2, \dots, B_n$ 是 S 的一个划分, 且 $P(B_j) > 0, j = 1, 2, \dots, n$, 则

$$P(A) = \sum_{j=1}^n P(B_j) P(A|B_j). \quad (1.4.6)$$

称 (1.4.6) 式为概率的全概率公式 (law of total probability).

$1, 2, \dots, n,$

$$P(B_k|A) = \frac{P(B_k A)}{P(A)} = \frac{P(B_k) P(A|B_k)}{\sum_{j=1}^n P(B_j) P(A|B_j)}. \quad (1.4.7)$$

称 (1.4.7) 式为概率的贝叶斯公式 (Bayes formula).

• 事件相互独立: 1. $P(AB) = P(A)P(B)$

$$2. P(B|A) = P(B)$$

定理 1.5.2 当事件 A 与事件 B 相互独立时, A 与 $\bar{B}, \bar{A}$ 与 $B, \bar{A}$ 与 $\bar{B}$ 均相互独立.

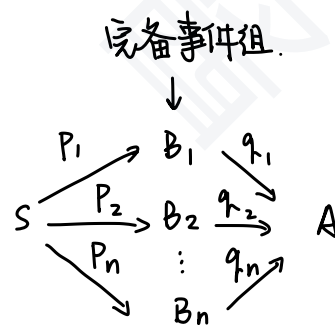
证明 因为 $A = AB \cup A\bar{B}$, 两边求概率并移项得

$$P(A\bar{B}) = P(A) - P(AB).$$

当事件 A 与事件 B 相互独立时,

$$P(A\bar{B}) = P(A) - P(A) \cdot P(B) = P(A) \cdot [1 - P(B)] = P(A) \cdot P(\bar{B}).$$

故 A 与 $\bar{B}$ 相互独立. 由此可进一步得 $\bar{A}$ 与 $B, \bar{A}$ 与 $\bar{B}$ 均相互独立.



定义 1.5.2 设 A, B, C 为三个随机事件, 当

$$P(AB) = P(A)P(B), P(AC) = P(A)P(C), P(BC) = P(B)P(C)$$

都成立时, 称事件 A, B, C 两两独立. 如果同时还满足

$$P(ABC) = P(A)P(B)P(C),$$

则称事件 A, B, C 相互独立.

应注意, 相互独立的事件一定是两两独立的, 而两两独立的事件不一定相互独立.

定义 1.5.3 设 n 个事件 $A_1, A_2, \dots, A_n (n \geq 2)$, 若对任意的 $k (2 \leq k \leq n)$, 都有

$$P(A_{i_1} A_{i_2} \cdots A_{i_k}) = \prod_{j=1}^k P(A_{i_j})$$

成立, 则称事件 $A_1, A_2, \dots, A_n (n \geq 2)$ 相互独立.

 例 1.5.3 一袋中有编号为 1, 2, 3, 4 共四个球, 从袋中有放回地取两次 (一次一个球, 假设取到每一个号码的概率相等), 并记录号码之和. 这样独立重复地试验, 求 “和等于 3” 出现在 “和等于 5” 之前的概率.

解 设 $A = \{\text{“和等于 3” 出现在 “和等于 5” 之前}\}$, $B = \{\text{第一次号码之和为 3}\}$, $C = \{\text{第一次号码之和为 5}\}$, $D = \{\text{第一次号码之和既不是 3, 也不是 5}\}$.

显然每次 “号码和为 3” 的概率为 $\frac{2}{16}$, “号码和为 5” 的概率为 $\frac{4}{16}$, “号码和既不是 3 又不是 5” 的概率为 $\frac{10}{16}$, 故

$$P(B) = \frac{2}{16}, \quad P(C) = \frac{4}{16}, \quad P(D) = \frac{10}{16}.$$

由全概率公式, 得

$$\begin{aligned} P(A) &= P(B) \cdot P(A|B) + P(C) \cdot P(A|C) + P(D) \cdot P(A|D) \\ &= \frac{2}{16} \times 1 + \frac{4}{16} \times 0 + \frac{10}{16} \times P(A|D). \end{aligned}$$

在已知第一次号码和既不是 3, 也不是 5 的条件下, 求 A 的条件概率问题, 相当于重新开始考虑 A 的概率, 因为试验是独立的, 即第一次结果不影响 A 发生的概率, 所以 $P(A|D) = P(A)$, 故

$$P(A) = \frac{2}{16} + \frac{10}{16} \times P(A),$$

从而

$$P(A) = \frac{1}{3}.$$

概率分布律 ①取值 ②取值对应概率

$$\Delta \sum_{k=0}^{+\infty} \frac{\lambda^k}{k!} = e^\lambda$$

1. 两点分布 $X \sim 0-1(p)$ 或 $B(1, p)$

L 分布律: $P(X=k) = p^k(1-p)^{1-k}, k=0, 1$

2. 二项分布 $X \sim B(n, p)$

$$P(X=k) = C_n^k p^k (1-p)^{n-k}, k=0, 1, 2, \dots$$

↑ A 在 n 重贝努利试验中发生 k 次

试验 E 仅有两个结果, E 独立重复 n 次.

$$\circ 1 = (p+q)^n = \sum_{k=0}^n C_n^k p^k (1-p)^{n-k}$$

3. 泊松分布 $X \sim P(\lambda)$

$$P(X=k) = \frac{\lambda^k e^{-\lambda}}{k!} \quad k=0, 1, 2, \dots \quad \lambda > 0$$

• $X \sim B(n, p)$ 与 $X \sim P(\lambda)$ 近似:

$$n > 0, p < 0.1 \text{ 时. } C_n^k p^k (1-p)^{n-k} \approx \frac{e^{-\lambda} \lambda^k}{k!}, \lambda = np$$

4. 超几何分布

$$P(X=k) = \frac{C_a^k C_b^{n-k}}{C_N^n} \quad k=l_1, l_1+1, \dots, l_2 \quad l_1 = \max(0, n-b) \quad l_2 = \min(a, n)$$

△不放回

5. 几何分布

$$P(X=k) = p(1-p)^{k-1}, k=1, 2, 3, \dots \quad 0 < p < 1$$

△出现某结果即停止试验.

6. 巴斯卡分布

$$P(X=k) = C_{k-1}^{r-1} p^r (1-p)^{k-r} \quad k=r, r+1, r+2, \dots$$

△某结果出现过 k 次即停止.

[随机变量 X 的分布函数] $F(x) = P\{X \leq x\}$

$$F(x) = P\{T \leq t\} = 1 - P\{T > t\}$$



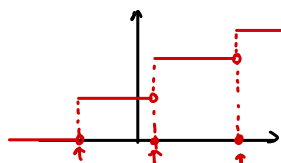
• $F(x+0) = F(x)$. 右连续

$F(x)$ 图象上跳跃的点 • $F(x) - F(x-0) = P(X=x)$

才会有在概率

跳跃的幅度

$$\circ F(-\infty) = 0 \quad F(+\infty) = 1$$



[连续型随机变量及概率密度函数]

X : 连续型随机变量

$F(x)$: X 的分布函数

$$F(x) = \int_{-\infty}^x f(t) dt$$

$f(x)$: X 的概率密度函数

$$\int_{-\infty}^{+\infty} f(x) dx = 1$$

$$P\{x_1 < X \leq x_2\} = F(x_2) - F(x_1)$$

$$\text{单点值 } P=0 = \int_{x_1}^{x_2} f(t) dt$$

$$f(x) = F'(x) = \lim_{\Delta x \rightarrow 0} \frac{P(x < X \leq x + \Delta x)}{\Delta x}$$

思考题:

设 A, B 为随机事件,

若 $P(A)=1$, 则 A 为必然事件吗?

若 $P(B)=0$, 则 B 为不可能事件吗?

若 $P(AB)=0$, 则 A 与 B 不相容吗?

答: 都不一定。例如:

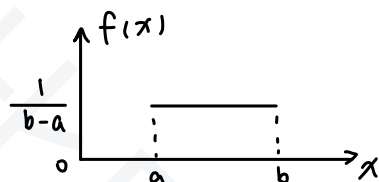
$X \in [0, 1], f(x) = \begin{cases} 1, & x \in [0, 1], \\ 0, & \text{其他} \end{cases}, A = \{0 < X < 1\},$

$B = \{X = 0.5\}$, 则 $P(A)=1, P(B)=0, P(AB)=0$.

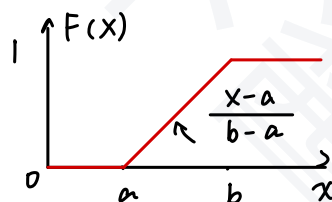
[几个重要的连续型随机变量分布]

1. 均匀分布 $X \sim U(a, b)$

$$f(x) = \begin{cases} \frac{1}{b-a}, & x \in (a, b) \\ 0, & \text{其他} \end{cases}$$



$\Rightarrow$

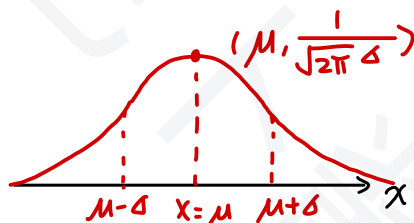


$$\Rightarrow \text{当 } a \leq c \leq c+1 \leq b, P(c < X < c+1) = \int_c^{c+1} \frac{1}{b-a} dt = \frac{1}{b-a}$$

2. 正态分布 $X \sim N(\mu, \sigma^2)$

密度 $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (\sigma > 0)$

分布: $\Phi = \int_{-\infty}^x f(t) dt$



$\sigma \downarrow$, 图形越高瘦.

μ : 位置参数, 决定对称轴位置.

σ : 尺度参数, 决定曲线分散性.

Δ [标准分布] $Z \sim N(0, 1)$

Z 的密度函数: $\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$

Z 的分布函数: $\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt \Rightarrow \Phi(x) + \Phi(-x) = 1$

$\Delta P(X \leq b) = \Phi(\frac{b-\mu}{\sigma}) \quad \Delta \text{当 } X \sim N(\mu, \sigma^2), \frac{X-\mu}{\sigma} \sim N(0, 1) \leftarrow \text{标准化}$

$\Delta P(|X-\mu| < b) = P(\frac{|X-\mu|}{\sigma} < \frac{b}{\sigma}) = \Phi(\frac{b}{\sigma}) - \Phi(-\frac{b}{\sigma})$

$P(X < a) = \Phi(\frac{a-\mu}{\sigma})$

3. 7

f

2

[随

定

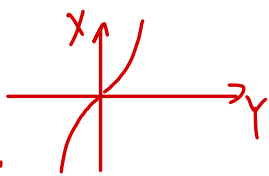
• 若 $X \sim N(\mu, \sigma^2)$, $Y = aX + b \Rightarrow Y \sim N(a\mu + b, a^2\sigma^2)$

• 设 $X \sim U(1)$ $F(x)$ 为 X 的分布函数. 记 $Y = F(X) \Rightarrow Y \sim U(0, 1)$

★ eg: $Y = \sin X$. $X \sim U(0, \pi)$ 求 $f_Y(y)$

$Y = \sin X$ 在 $(0, \pi)$ 不单调

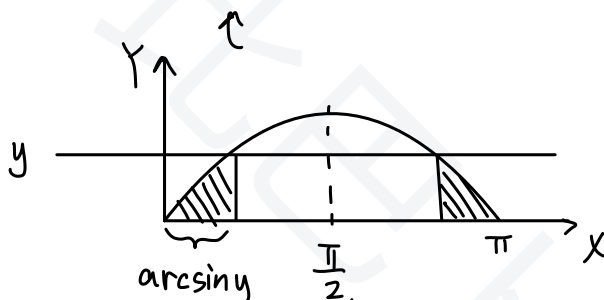
$X = \arcsin Y$



对称性. ΔX 长度一致
 $P(\Delta X)$ 一致

$\therefore$ 对于 $0 < y \leq 1$: $F_Y(y) = P(Y \leq y) = P(\sin X \leq y) = \frac{2 \arcsin y}{\pi}$ (概率率之和, 因为 X 均匀分布)

$$\therefore f_Y(y) = \begin{cases} \frac{2}{\pi \sqrt{1-y^2}} & 0 < y < 1 \\ 0 & \text{其他} \end{cases}$$



$$P(A \cup B) = P(A) + P(B) - P(AB)$$

$$P(A|B) = P(AB)/P(B) \rightarrow P(AB) = P(B)P(A|B) = P(A)P(B|A)$$

$$P(AB|C) = P(A|C)P(B|AC)$$

$$\text{全概率率 } P(A) = \sum_{j=1}^n P(B_j)P(A|B_j)$$

$$A, B \text{ 独立} \rightarrow P(B) = P(B|A), P(AB) = P(A)P(B)$$

$$X \sim B(n, p) \quad P(X=k) = C_n^k p^k (1-p)^{n-k} \quad \text{二项分布, } n \text{ 重伯努利}$$

$$X \sim P(\lambda) \quad P(X=k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

$$\text{几何分布: 出现结果停止. } P = p(1-p)^{k-1} \quad (X=k)$$

$$\text{帕斯卡分布, 出现结果 } r \text{ 次停止. } P(X=k) = C_{k-1}^{r-1} p^r (1-p)^{k-r}$$

恰在第 k 次试验中, 结果出现了 r 次.

$$F(x) = P(X \leq x) \quad F(x) \text{ 右连续.}$$

$$= \int_{-\infty}^x f(t) dt \rightarrow f(t) = F'(t)$$

$$P\{a \leq X \leq b\} = F(b) - F(a).$$

$$\text{均匀分布: } f(x) = \frac{1}{b-a} \quad (x \in (a, b))$$

$$X \sim U(a, b).$$

$$\text{正态分布} \quad f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \quad (\sigma > 0)$$

$$X \sim N(\mu, \sigma^2), \quad \Phi = \int_{-\infty}^x f(t) dt \rightarrow P(X \leq b) = \Phi\left(\frac{b-\mu}{\sigma}\right).$$

$$Z \sim N(0, 1) \quad \varphi(z) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{z^2}{2}\right], \quad \Phi(x) + \Phi(-x) = 1.$$

指数分布.

$$X \sim E(\lambda) \quad f(x) = \lambda e^{-\lambda x} \quad (x > 0) \quad F(x) = 1 - e^{-\lambda x} \quad (x > 0)$$

$$X \sim f_X(x), \quad -\infty < x < +\infty. \quad Y = g(X), \quad X = h(Y), \quad g'(x) > 0 / < 0$$

$$\Rightarrow f_Y(y) = f_X(h(y)) |h'(y)|, \quad (\alpha < y < \beta).$$

第三章 多元随机变量及其分布.

[二元离散型随机变量] 样本空间 $S = \{e\}$, 设 $X = X(e)$ 和 $Y = Y(e)$ 是定义在 S 上的随机变量. 称其构成的向量 (X, Y) 为二元(维)随机变量.

[联合概率分布]

1. 离散型随机变量: (X, Y) 全部取值可列

2. 离散型随机变量的联合概率分布律.

$X \backslash Y$	y_1	y_2	y_3	$\dots$
x_1				
x_2				
x_3				
$\vdots$				

[边际分布]

对离散型随机变量 (X, Y) , 联合分布律: $P(X=x_i, Y=y_j) = P_{ij}$, $i, j = 1, 2, \dots$

则 X 的边际分布律: $P(X=x_i) = P(X=x_i, Y < +\infty) = \sum_{j=1}^{\infty} P_{ij} \stackrel{\text{def}}{=} P_{i\cdot}$, $i = 1, 2, \dots$

Y 的边际分布律: $P(Y=y_j) = P(X < +\infty, Y=y_j) = \sum_{i=1}^{\infty} P_{ij} \stackrel{\text{def}}{=} P_{\cdot j}$, $j = 1, 2, \dots$

[条件分布] 对离散型随机变量 (X, Y) (前提: 条件的 $p > 0$)

1. $\{Y=y_j\}$ 条件下, 随机变量 X 的条件分布律: $P(X=x_i | Y=y_j) = \frac{P(X=x_i, Y=y_j)}{P(Y=y_j)} = \frac{P_{ij}}{P_{\cdot j}}$

$\hookrightarrow$ 不同 x_i 取值对应 p 不同

2. $\{X=x_i\}$ 条件下, 随机变量 Y 的条件分布律: $P(Y=y_j | X=x_i) = \frac{P(X=x_i, Y=y_j)}{P(X=x_i)} = \frac{P_{ij}}{P_{i\cdot}}$

二元随机变量的分布函数.

(一) 联合分布函数.

(X, Y) 为二元随机变量, $F(x, y) = P\{(X \leq x) \cap (Y \leq y)\} = P(X \leq x, Y \leq y)$

称 $F(x, y)$ 为 (X, Y) 的联合分布函数.

1° $F(x, y)$ 关于 x, y 单调不减, 关于 x, y 右连续.

2° $0 \leq F(x, y) \leq 1$, $F(+\infty, +\infty) = 1$, $F(-\infty, y) = F(x, -\infty) = F(-\infty, -\infty) = 0$

3° $x_1 < x_2, y_1 < y_2 \Rightarrow F(x_2, y_2) - F(x_2, y_1) - F(x_1, y_2) + F(x_1, y_1) \geq 0$

(二) 边际分布函数.

$F_X(x) = F(x, +\infty) = P(X \leq x) = P(X \leq x, Y \leq +\infty)$

$F_Y(y) = F(+\infty, y)$

(三) 条件分布函数.

$$1^\circ P(Y=y) > 0$$

$$F_{X|Y}(x|y) = P(X \leq x | Y=y) = \frac{P(X \leq x, Y=y)}{P(Y=y)}$$

$$2^\circ P(Y=y)=0. \text{ 但对 } \forall \varepsilon > 0, P(y < Y \leq y+\varepsilon) > 0$$

$$F_{X|Y}(x|y) = \lim_{\varepsilon \rightarrow 0^+} P(X \leq x | y < Y \leq y+\varepsilon) = \lim_{\varepsilon \rightarrow 0^+} \frac{P(X \leq x, y < Y \leq y+\varepsilon)}{P(y < Y \leq y+\varepsilon)}$$

[二元连续型随机变量]

(一) 联合概率密度函数.

对 (X, Y) 的分布函数 $F(x, y)$ 存在非负 $f(x, y)$, 满足

$$F(x, y) = \int_{-\infty}^y \int_{-\infty}^x f(u, v) du dv$$

称 $(X, Y) \leftarrow$ 二元连续型随机变量

$f(x, y)$ 联合概率密度函数.

$$1^\circ f(x, y) \geq 0$$

$$2^\circ \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy = 1$$

$$3^\circ P\{(X, Y) \in G\} = \iint_G f(x, y) dx dy$$

$$4^\circ \text{ 在 } f(x, y) \text{ 连续点有 } f(x, y) = \frac{\partial^2 F(x, y)}{\partial x \partial y}$$

算=重积分.

• $z = f(x, y)$ 表示空间曲面. z 与 XOY 体积=1

• $P\{(X, Y) \in G\}$ 表以 G 为底. z 为顶的曲面圆柱体积.

(二) 边际概率密度函数.

$$F_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \int_{-\infty}^x f_X(u) du$$

$$F_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx = \int_{-\infty}^y f_Y(v) dv$$

$$\text{eg: } f(x, y) = \begin{cases} 15y, & 0 < x < 1, x^2 < y < x \\ 0, & \text{其他.} \end{cases}$$

$$\Rightarrow f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \int_{x^2}^x 15y dy = \frac{15}{2}(x^2 - x^4), \quad 0 < x < 1; \quad 0, \text{其他}$$

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx = \int_y^{\sqrt{y}} 15y dx = 15(y^{\frac{3}{2}} - y^2), \quad 0 < y < 1; \quad 0, \text{其他.}$$

(三) 条件概率密度函数.

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}, \quad f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)}$$

$$\Rightarrow F_{X|Y}(x|y) = \int_{-\infty}^x \frac{f(u, y)}{f_Y(y)} du$$

$$F_{Y|X}(y|x) = \int_{-\infty}^y \frac{f(x, v)}{f_X(x)} dv$$

$$\begin{aligned} F_{X|Y}(x|y) &= \lim_{\Delta y \rightarrow 0^+} \frac{P(X \leq x, y < Y \leq y + \Delta y)}{P(y < Y \leq y + \Delta y)} \\ &= \lim_{\Delta y \rightarrow 0^+} \frac{\frac{1}{\Delta y} \cdot \int_{-\infty}^x ds \int_y^{y+\Delta y} f(u, v) dv}{\frac{1}{\Delta y} \cdot \int_y^{y+\Delta y} f_Y(t) dt} \\ &= \int_{-\infty}^x \frac{f(u, y)}{f_Y(y)} du. \end{aligned}$$

$$1^\circ f_{X|Y}(x|y) \geq 0$$

$$2^\circ \int_{-\infty}^{+\infty} f_{X|Y}(x|y) dx = 1$$

$$3^\circ P\{a < X < b | Y=y\} = \int_a^b f_{X|Y}(x, y) dx$$

$$4^\circ f_{X|Y}(x|y) \text{ 连续点: } f_{X|Y}(x|y) = \frac{dF_{X|Y}(x|y)}{dy}$$

$$5^\circ f(x, y) = f_{X|Y}(x|y) f_Y(y) = f_{Y|X}(y|x) \cdot f_X(x)$$

(四) 二元均匀分布和二元正态分布.

(1) 均匀分布 $(X, Y) \in D$

$$f(x, y) = \begin{cases} \frac{1}{D \text{ 的面积}}, & (x, y) \in D \\ 0, & \text{其他} \end{cases}$$

$$\text{若 } D_1 \subset D, \quad P\{(X, Y) \in D_1\} = \frac{D_1 \text{ 的面积}}{D \text{ 的面积}}$$

(2) 二元正态分布 $(X, Y) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$

$$\hookrightarrow f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \times \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_1)^2}{\sigma_1^2} - 2\rho\frac{(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \frac{(y-\mu_2)^2}{\sigma_2^2}\right]\right\}$$

$$1. f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{(x-\mu_1)^2}{2\sigma_1^2}}, \text{ 即 } X \sim N(\mu_1, \sigma_1^2)$$

二元正态分布的边缘分布是正态分布且不依赖于 ρ

$$2. f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)}$$

$\Leftrightarrow$ 在 $\{X=x\}$ 条件下, Y 的条件分布即 $Y \sim N(\mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x-\mu_1), (1-\rho^2)\sigma_2^2)$

在 $\{Y=y\}$ 条件下, X 的条件分布即 $X \sim N(\mu_1 + \rho\frac{\sigma_1}{\sigma_2}(y-\mu_2), (1-\rho^2)\sigma_1^2)$

$$\int_{-\infty}^{+\infty} e^{-(y^2+y)} dy = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi} \cdot \sqrt{\frac{1}{2}}} e^{-\frac{(y+\frac{1}{2})^2}{2 \cdot \frac{1}{2}}} \cdot \sqrt{\pi} \cdot e^{\frac{1}{4}} = \sqrt{\pi} \cdot e^{\frac{1}{4}}$$

正态分布密度函数

先把被积函数整理为正态分布的形式. 利用 $\int_{-\infty}^{+\infty}$ 密度函数 = 1 代简

$$X \sim N(\mu, \sigma^2) \Rightarrow f(x) = \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

[随机变量独立性]

$F(x, y)$: (X, Y) 的联合分布函数.

$F_X(x), F_Y(y)$: (X, Y) 的边缘分布函数.

$\Rightarrow$ 若 $F(x, y) = F_X(x)F_Y(y)$ 则称随机变量 X, Y 相互独立

1. (X, Y) 离散型: $P_{ij} = p_i \cdot p_j$

2. (X, Y) 连续型: 注意是密度函数 $f(x, y) = f_X(x) \cdot f_Y(y)$ [除零面积集以外处处成立]

Δ 对于二维正态随机变量 (X, Y) , X 与 Y 相互独立的充要条件: $\rho = 0$

* " $\Rightarrow$ " 若 X, Y 独立. 因 $f(x, y), f_X(x), f_Y(y)$ 连续 $\Rightarrow \forall x, y$, 有: $f(x, y) = f_X(x)f_Y(y)$

$$\Rightarrow X = \mu_1, Y = \mu_2 \text{ 时, } f(\mu_1, \mu_2) = f_X(\mu_1) \cdot f_Y(\mu_2) \Rightarrow \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} = \frac{1}{2\pi\sigma_1\sigma_2} \Rightarrow \rho = 0$$

[一般 n 元随机变量]

1. n 元随机变量: n 元向量 $(X_1, X_2, \dots, X_n)$

2. 分布函数: $F(X_1, X_2, \dots, X_n) = P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n)$

3. 分布律

4. 概率密度函数 $f(x_1, x_2, \dots, x_n) = \int_{-\infty}^{x_n} \int_{-\infty}^{x_{n-1}} \dots \int_{-\infty}^{x_1} f(x_1, x_2, \dots, x_n) dx_1 dx_2 \dots dx_n$

5. 边缘分布.

$$F_{X_1}(x_1) = F(x_1, \infty, \infty, \dots, \infty)$$

$$F_{(X_1, X_2)}(x_1, x_2) = F(x_1, x_2, \infty, \dots, \infty)$$

$$P(X_1 = x_{1i_1}) = \sum_{i_2, i_3, \dots, i_n} P(X_1 = x_{1i_1}, X_2 = x_{2i_2}, \dots, X_n = x_{ni_n})$$

$$P(X_1 = x_{1i_1}, X_2 = x_{2i_2}) = \sum_{i_3, i_4, \dots, i_n} P(X_1 = x_{1i_1}, X_2 = x_{2i_2}, \dots, X_n = x_{ni_n})$$

$$f_{X_1}(x_1) = \int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} f(x_1, x_2, \dots, x_n) dx_2 dx_3 \dots dx_n$$

$$f_{(X_1, X_2)}(x_1, x_2) = \int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} f(x_1, x_2, \dots, x_n) dx_3 dx_4 \dots dx_n$$

6. 独立性

(1) $X_1, X_2, \dots, X_n$ 相互独立: $F(x_1, x_2, \dots, x_n) = F_{X_1}(x_1) F_{X_2}(x_2) \dots F_{X_n}(x_n)$

(2) $(X_1, X_2, \dots, X_m)$ 与 $(Y_1, Y_2, \dots, Y_n)$ 独立性: $F(x_1, x_2, \dots, x_m, y_1, y_2, \dots, y_n) = F(x_1, x_2, \dots, x_m) \cdot F(y_1, y_2, \dots, y_n)$

◎ 定理

设 $(X_1, X_2, \dots, X_m)$ 与 $(Y_1, Y_2, \dots, Y_n)$ 相互独立,

则 $X_i (i = 1, 2, \dots, m)$ 与 $Y_j (j = 1, 2, \dots, n)$ 相互独立;

若 $h(x_1, x_2, \dots, x_m)$ 和 $g(y_1, y_2, \dots, y_n)$ 是连续函数,

则 $h(X_1, X_2, \dots, X_m)$ 和 $g(Y_1, Y_2, \dots, Y_n)$ 相互独立.

[二元随机变量的函数分布]

$$[Z = X + Y]$$

(1).

$$P(Z = z_k) = \sum_{i=1}^{+\infty} P(X = x_i, Y = z_k - x_i), \quad k=1, 2, \dots$$

$$= \sum_{j=1}^{+\infty} P(X = z_k - y_j, Y = y_j), \quad k=1, 2, \dots$$

$$X, Y \text{ 独立时, } P(Z = z_k) = \sum_{i=1}^{+\infty} P(X = x_i)P(Y = z_k - x_i), \quad k=1, 2, \dots$$

(2). Z 的密度函数.

$$f_Z(z) = \int_{-\infty}^{+\infty} f(z-y, y) dy = \int_{-\infty}^{+\infty} f(x, z-x) dx$$

(3). 卷积公式:

$$X, Y \text{ 相互独立时, } f_Z(z) = \int_{-\infty}^{+\infty} f_X(z-y)f_Y(y) dy = \int_{-\infty}^{+\infty} \underbrace{f_X(x)}_{\text{用 } y=z-x \text{ 换元 } \int \text{再对 } dx \text{ 积分}} f_Y(z-x) dx$$

例5.4 设X和Y是相互独立的标准正态随机变量, 求 $Z = X + Y$ 的概率密度函数.

$$f_Z(z) = \int_{-\infty}^{+\infty} f_X(x)f_Y(z-x)dx = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{-\frac{x^2}{2}} e^{-\frac{(z-x)^2}{2}} dx$$

$$= \frac{1}{2\sqrt{\pi}} e^{-\frac{z^2}{4}} \int_{-\infty}^{+\infty} \frac{1}{\sqrt{\pi}} e^{-\frac{(x-\frac{z}{2})^2}{2}} dx = \frac{1}{2\sqrt{\pi}} e^{-\frac{z^2}{4}}$$

即 $Z \sim N(0, 2)$

一般地, 设X与Y相互独立,

$$X \sim N(\mu_1, \sigma_1^2), Y \sim N(\mu_2, \sigma_2^2)$$

$$\text{则 } Z = X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

$$\text{结合 } aX \sim N(a\mu_1, a^2\sigma_1^2),$$

$$bY + c \sim N(b\mu_2 + c, b^2\sigma_2^2), \text{ 得}$$

$$aX + bY + c \sim N(a\mu_1 + b\mu_2 + c, a^2\sigma_1^2 + b^2\sigma_2^2)$$

例5.3 设随机变量 $X \sim P(\lambda_1), Y \sim P(\lambda_2)$,

且X, Y相互独立. 若 $Z = X + Y$,

求Z的概率分布律.

$$\text{解: } P(X = i) = \frac{\lambda_1^i e^{-\lambda_1}}{i!}, \quad i = 0, 1, 2, \dots,$$

$$P(Y = j) = \frac{\lambda_2^j e^{-\lambda_2}}{j!}, \quad j = 0, 1, 2, \dots$$

$$P(Z = k) = \sum_{i=0}^k P(X = i)P(Y = k-i) = \sum_{i=0}^k \frac{\lambda_1^i e^{-\lambda_1}}{i!} \frac{\lambda_2^{k-i} e^{-\lambda_2}}{(k-i)!}$$

$$= \frac{e^{-(\lambda_1 + \lambda_2)}}{k!} \sum_{i=0}^k \frac{k!}{i!(k-i)!} \lambda_1^i \lambda_2^{k-i}$$

$$= \frac{(\lambda_1 + \lambda_2)^k e^{-(\lambda_1 + \lambda_2)}}{k!}, \quad k = 0, 1, 2, \dots$$

$$\text{即 } Z = X + Y \sim P(\lambda_1 + \lambda_2).$$

用 $y = z - x$ 换元 $\int$ 再对 dx 积分

$$\int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi} \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} d\sqrt{2}x$$

$$= \sqrt{\pi} \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt = \sqrt{\pi}$$

$$[M = \max\{X, Y\}, N = \min\{X, Y\}]$$

$$M \text{ 的分布: } F_{\max}(Z) = P(X \leq Z) P(Y \leq Z)$$

$$= F_X(Z) F_Y(Z)$$

$$N \text{ 的分布: } F_{\min}(Z) = 1 - P(N > Z)$$

$$= 1 - P(X > Z) P(Y > Z)$$

$$= 1 - (1 - F_X(Z))(1 - F_Y(Z))$$

第四章：随机变量的数字特征

[数学期望]

1. 离散型：分布律 $P(X=x_k)=p_k, k=1,2,\dots$ ，若 $\sum_{k=1}^{+\infty} |x_k| p_k < \infty$ 。

$$\text{则 } E(X) = \sum_{k=1}^{+\infty} x_k p_k.$$

2. 连续型：概率密度 $f(x)$ 若 $\int_{-\infty}^{+\infty} |x| f(x) dx < \infty$ 。

$$\text{则 } E(X) = \int_{-\infty}^{+\infty} x f(x) dx$$

eg. $X \sim P(\lambda)$. 则 $E(X) = \lambda$.

$$\Rightarrow E(X) = \sum_{k=0}^{+\infty} k \frac{\lambda^k e^{-\lambda}}{k!} = \lambda e^{-\lambda} \sum_{k=1}^{+\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda} \cdot e^{\lambda} = \lambda$$

eg. $X \sim E(\lambda)$. 则 $E(X) = \frac{1}{\lambda}$

eg. 若 $X_1, X_2, \dots, X_5$ 独立且同分布。

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0 \\ 0, & x \leq 0 \end{cases} \quad F(x) = \begin{cases} 1 - e^{-\lambda x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

令 $N = \min\{X_1, X_2, \dots, X_5\}$, $M = \max\{X_1, X_2, \dots, X_5\}$ 求 $E(N)$, $E(M)$

$$\textcircled{1} F_N(y) = P\{N \leq y\} = 1 - P\{N > y\} = 1 - P\left\{\bigcap_{i=1}^5 (X_i > y)\right\} = 1 - [1 - F(y)]^5$$

$$= \begin{cases} 1 - e^{-5\lambda y}, & y > 0 \\ 0, & \text{其他} \end{cases} \Rightarrow f_N(y) = \begin{cases} 5\lambda e^{-5\lambda y}, & y > 0 \\ 0, & \text{其他} \end{cases}$$

$$\textcircled{2} F_M(y) = P\{M \leq y\} = P\left\{\bigcap_{i=1}^5 (X_i \leq y)\right\} = [F(y)]^5$$

$$= \begin{cases} (1 - e^{-\lambda y})^5, & y > 0 \\ 0, & \text{其他} \end{cases} \Rightarrow f_M(y) = \begin{cases} 5\lambda e^{-\lambda y} (1 - e^{-\lambda y})^4, & y > 0 \\ 0, & \text{其他} \end{cases}$$

[随机变量函数的数学期望]

连续函数 $Y = g(X)$:

1. X 离散型. $\sum_{k=1}^{+\infty} |g(x_k)| p_k < \infty$, $E(Y) = E[g(X)] = \sum_{k=1}^{+\infty} g(x_k) p_k$.

2. X 连续型. $\int_{-\infty}^{+\infty} |g(x)| f(x) dx < \infty$, $E(Y) = \int_{-\infty}^{+\infty} g(x) f(x) dx$

二元连续函数 $Z = h(X, Y)$

1. (X, Y) 离散. $E(Z) = \sum_{i=1}^{+\infty} \sum_{j=1}^{+\infty} h(x_i, y_j) p_{ij}$

2. (X, Y) 连续. $E(Z) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h(x, y) f(x, y) dx dy$

$$\text{eg. } E(X^2) = \int_{-\infty}^{+\infty} x^2 f(x) dx, \quad E(XY) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy f(x, y) dx dy.$$

数学期望的特性.

$$1. E(C) = C$$

$$2. E(CX) = CE(X)$$

$$3. E(C_0 + \sum_{i=1}^n C_i X_i) = C_0 + \sum_{i=1}^n C_i E(X_i)$$

$$\sum_{i=1}^n C_i X_i = \sum_{i=1}^n C_i E(X_i)$$

4. 相互独立的随机变量:

$$\sum_{i=1}^n (\prod_{i=1}^n X_i) = \prod_{i=1}^n E(X_i)$$

$$[方差] \quad D(X) / Var(X) = E\{[X - E(X)]^2\}$$

$$[标准差/均方差] \quad \Delta(X) = \sqrt{Var(X)}$$

$$1. \text{离散型, } Var(X) = \sum_{i=1}^{\infty} [X_i - E(X)]^2 p_i$$

$$2. \text{连续型, } Var(X) = \int_{-\infty}^{+\infty} [X - E(X)]^2 f(x) dx$$

$$[方差计算公式]: Var(X) = E(X^2) - [E(X)]^2$$

$$= E\{[X - E(X)]^2\} = E\{X^2 - 2XE(X) + [E(X)]^2\} = E(X^2) - 2E(X)E(X) + [E(X)]^2$$

$$(1) 0-1 \text{ 分布: } Var(X) = p(1-p)$$

$$(2) X \sim P(\lambda): Var(X) = E(X) = \lambda$$

$$(3) X \sim U(a, b): Var(X) = \frac{(b-a)^2}{12}$$

$$(4) X \sim E(\lambda): Var(X) = \frac{1}{\lambda^2} \quad E(X) = \frac{1}{\lambda}$$

$$(5) X \sim B(n, p): E(X) = np, Var(X) = np(1-p)$$

$$(6) X \sim N(\mu, \sigma^2): E(X) = \mu, Var(X) = \sigma^2$$

$$\rightarrow C_0 + C_1 X_1 + C_2 X_2 + \dots \sim N(C_0 + C_1 \mu_1 + C_2 \mu_2 + \dots, C_1^2 \sigma_1^2 + C_2^2 \sigma_2^2 + \dots)$$

[方差的性质]:

$$1. Var(C) = 0$$

$$2. Var(CX) = C^2 Var(X)$$

$$3. Var(X+Y) = Var(X) + Var(Y) + 2E\{[X - E(X)][Y - E(Y)]\}$$

$$\text{当 } X, Y \text{ 相互独立, 则 } Var(aX + bY + C) = a^2 Var(X) + b^2 Var(Y)$$

$$4. Var(X) = 0 \Leftrightarrow P(X=C) = 1 \text{ 且 } E(X) = C$$

[标准变量 X^*]: 若随机变量 X 有: $E(X) = \mu$, $Var(X) = \sigma^2$.

$$\text{则 } X^* = \frac{X - \mu}{\sigma}, \Rightarrow E(X^*) = 0, Var(X^*) = 1$$

[协方差与相关系数]

$$[\text{协方差}] Cov(X, Y) = E\{[X - E(X)][Y - E(Y)]\}$$

$$- Cov(X, Y) = E(XY) - E(X)E(Y)$$

$$= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} (x - E(X))(y - E(Y)) f(x, y) dx dy.$$

$$= \sum_{i=1}^{+\infty} \sum_{j=1}^{+\infty} (x_i - E(X))(y_j - E(Y)) P_{ij}$$

$$- Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$$

[协方差性质] 1. $Cov(X, Y) = Cov(Y, X)$

$$2. Cov(X, X) = Var(X)$$

$$3. Cov(aX, bY) = abCov(X, Y)$$

$$4. Cov(X_1 + X_2, Y) = Cov(X_1, Y) + Cov(X_2, Y)$$

$$5. Var(X)Var(Y) \neq 0 \text{ 时, } (Cov(X, Y))^2 \leq Var(X)Var(Y).$$

等号当且仅当 $P(Y = a + bX) = 1$ 时取到.

$$b. Cov(aX + bY, cX + dY) = acVar(X) + bdVar(Y) + (ad + bc)Cov(X, Y)$$

$$7. Var\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n Var(X_i) + 2 \sum_{1 \leq i < j \leq n} Cov(X_i, X_j)$$

$$[\text{相关系数}] \rho_{xy} = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}} \quad \rightsquigarrow Cov(X, Y) = 0 \Leftrightarrow X, Y \text{ 不相关.}$$

$$= Cov\left(\frac{X - E(X)}{\sqrt{Var(X)}}, \frac{Y - E(Y)}{\sqrt{Var(Y)}}\right)$$

[相关系数性质]

$$1. |\rho_{xy}| \leq 1$$

$$2. |\rho_{xy}| = 1 \Leftrightarrow P(Y = a + bX) = 1; \rho_{xy} = 1, b > 0, \rho_{xy} = -1, b < 0$$

$$3. \rho_{xy} = 0 \Leftrightarrow Cov(X, Y) = 0 \Leftrightarrow E(XY) = E(X)E(Y) \Leftrightarrow Var(X + Y) = Var(X) + Var(Y)$$

$\Delta X, Y$ 相互独立 $\Rightarrow X, Y$ 不相关; X, Y 不相关 $\nRightarrow X, Y$ 相互独立.

[均匀方误差] $e(a, b) = E\{[Y - (a + bX)]^2\}$. 衡量线性近似度.

$$\min_{a, b} e(a, b) = e(a_0, b_0)$$

$$\Rightarrow \begin{cases} a_0 = E(Y) - b_0 E(X) \\ b_0 = \frac{Cov(X, Y)}{Var(X)} \end{cases},$$

$$e(a_0, b_0) = E\{[Y - (a + bX)]^2\} = (1 - \rho_{xy}^2) Var(Y)$$

$$(X, Y) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$$

eg: (X, Y) 服从二元正态分布, $f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}$

$$\Rightarrow \text{Cov}(X, Y) = \rho\sigma_1\sigma_2$$

$$\rho_{XY} = \rho$$

$\rightarrow X$ 和 Y 相互独立 $\Leftrightarrow \rho = 0 \Leftrightarrow \rho_{XY} = 0 \Leftrightarrow X, Y$ 不相关.

4.4 其它数字特征

定义: 设 X 和 Y 是随机变量

若 $E(X^k)$ $k=1, 2, \dots$ 存在,

则称它为 X 的 k 阶(原点)矩;

若 $E\{[X - E(X)]^k\}$ $k=1, 2, \dots$ 存在,

则称它为 X 的 k 阶中心矩;

若 $E\{X^k Y^l\}$ 存在 $k, l=1, 2, \dots$ 存在,

则称它为 X 和 Y 的 $k+l$ 阶混合(原点)矩;

若 $E\{[X - E(X)]^k [Y - E(Y)]^l\}$ $k, l=1, 2, \dots$ 存在,

则称它为 X, Y 的 $k+l$ 阶混合中心矩;

[上分位数]

定义: X 为连续型随机变量, 其分布函数和概率密度函数分别为 $F(x)$ 和 $f(x)$, 称满足条件

$$P\{X > x_\alpha\} = 1 - F(x_\alpha) = \int_{x_\alpha}^{+\infty} f(x)dx = \alpha$$

的实数 x_α 为随机变量 X (或此分布) 的 上(侧) α 分位数.

特别地, 当 $\alpha = 1/2$ 时, $x_{1/2}$ 称为 X 的中位数;

当 $\alpha = 1/4$ 时, $x_{1/4}$ 称为 X 的上 $1/4$ 分位数;

当 $\alpha = 3/4$ 时, $x_{3/4}$ 称为 X 的上 $3/4$ 分位数.

设 $X \sim N(0, 1)$, 若 z_α 满足条件 $P\{X > z_\alpha\} = \alpha, 0 < \alpha < 1$

则称点 z_α 为标准正态分布的上 α 分位数.

